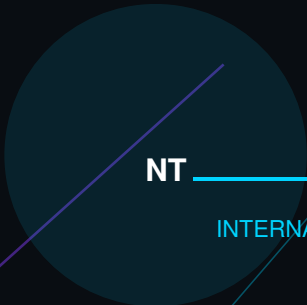




Research Notes

Problem framing, goal association, assumptions and associated research papers



PROJECT
Node v0

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Research Question
Which motor housing architecture will fulfill NT-N-G1, NT-N-G2 and NT-N-G7?

Abstract
This document collates and analyses multiple research papers and notes for fulfilling Node v0 design goals. Its intended purpose is to isolate observations, assumptions, risks, and design direction before committing to a detailed technical design.

Keywords: integrated actuator, mechanical design, motor housing, robotics, research notes

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1 RESEARCH SNAPSHOT

Research Snapshot

Problem	Build motor housing to maximise thermal and electromagnetic performance of the actuator, while reducing size and weight.
Key Finding	The highest leverage early decision is the actuator architecture, because it constrains torque density, controllability, thermal behaviour, and manufacturability.
Main Uncertainty	The true operating envelope is still uncertain: load profile, duty cycle, heat rejection, encoder resolution, and gearbox backlash all need validation.
Next Step	Prototype the smallest credible bench rig and measure torque, speed, temperature rise, backlash, and closed-loop response.

2 PROBLEM FRAMING

Research Question – System Direction

What outrunner BLDC motor housing architectures minimises thermal resistance ($^{\circ}\text{C}/\text{W}$), magnetic flux leakage and Air Gap instability ($\pm \text{mm}$)?

3 PRELIMINARY DIRECTION

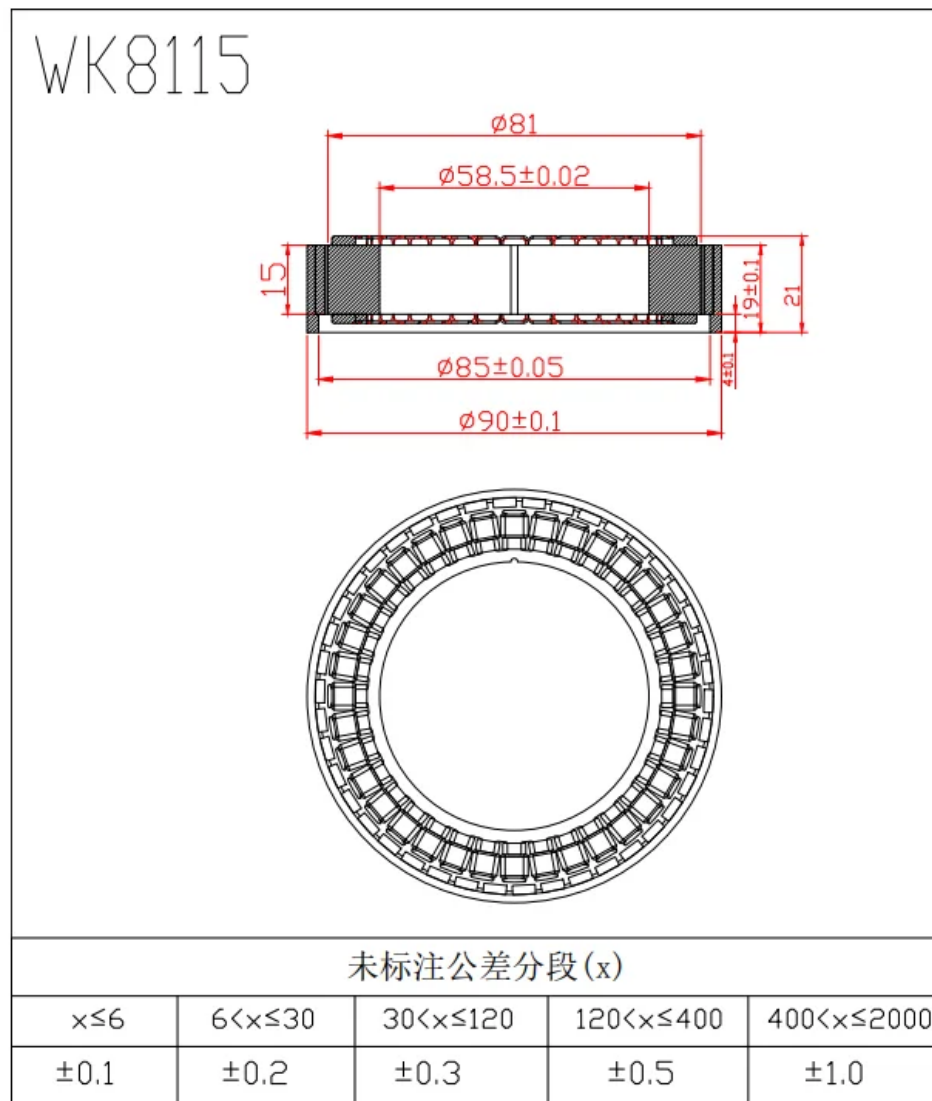
This research document will collate results from multiple research papers and notes to inform the design of a motor housing for Node v0. The goal is to identify the most promising motor housing architecture that meets the specified design goals (NT-N-G1, NT-N-G2, NT-N-G7). A preliminary frameless outrunner BLDC motor has been selected, informed by cost and performance requirements.

4 ASSUMPTIONS

- The frameless outrunner BLDC architecture will provide the highest torque density and lowest thermal resistance for the Node v0 actuator.
- Aluminium provides the best balance of thermal conductivity, weight, and manufacturability for the motor housing.

5 FRAMELESS BLDC ARCHITECTURE

The selected frameless outrunner BLDC was the SteadyWin WK8115 model with the following dimensions:



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Figure 1: SteadyWin WK8115 Dimensions

The magnetic, thermal and electrical properties of the motor are taken from the manufacturer as follows:

规格型号	wk1105	wk2205	wk2806	wk3505	wk3510	wk4310	wk4315	wk5215	wk5225	wk6010	wk6025	wk8110	wk8115	wk10025
额定电压 (V)	12	12	12	12	12	16	16	24	24	24	36	24	48	48
额定电流 (A)	0.59	0.66	0.81	0.79	0.53	0.79	0.59	1.12	1.28	1.19	1.78	1.75	6.5	4
堵转电流 (A)	0.92	1	1.22	1.2	0.8	1.2	0.9	1.69	1.94	1.8	2.67	3	10	11.2
额定扭矩 (N.M)	0.01	0.04	0.06	0.07	0.11	0.21	0.28	0.76	1.26	0.73	1.77	1.6	2.16	6
堵转扭矩 (N.M)	0.02	0.07	0.1	0.12	0.19	0.32	0.42	1.24	1.92	1.21	2.83	2.68	5.83	8.9
额定转速 (rpm)	7000	5500	2060	1890	945	585	425	390	313	280	227	186	1800	190
最高转速 (rpm)	8700	7200	2570	2160	1250	875	608	565	455	410	315	312	2280	310
绕组匝数 (T)	50	65	60	40	160	160	160	50	80	50	30	30	30	20
相间电阻 (Ω)	12.5	8	5	6.5	8.8	10.9	15.2	14.8	11.8	11.9	10.1	3	0.4	2.5
相间电感 (mH)	1.15	1.45	1.51	2.35	3.95	4.74	6.56	7.8	11.31	13.03	10.07	31.5	0.2	66
转速常数 (rpm/V)	108	100	121	74	45	33	23	12	12	11	7	11	37	12
扭矩常数 (N.M/A)	0.01	0.06	0.07	0.09	0.2	0.27	0.47	0.68	0.98	0.61	1.63	1.13	0.58	0.76
转子惯量 (gcm ²)	15	42	51	56	81	202	268	667	933	748	1645	2860	1500	4656
极对数	6	7	7	11	11	14	14	11	11	14	14	21	21	21
工作温度范围 (℃)	-20~80													
最高退磁温度 (℃)	120													

Figure 2: SteadyWin WK8115 Specifications

Figure 1 will inform the tolerances and mechanical design of the Node v0 motor housing, while Figure 2 will inform the thermal and electromagnetic characteristics of the Node v0 actuator.

6 KOLLMORGEN MOUNTING AND INSTALLATION GUIDELINES

Kollmorgen is a German engineering company that provides a range of motion control solutions. While the procured SteadyWin WK8115 is not a Kollmorgen product, the company provides a set of mounting and installation guidelines that can be applied to the Node v0 motor housing design.

6.0.1 Bearings

Bearings in the motor housing are critical for maintaining air gap stability and therefore electromagnetic performance [1]. Kollmorgen notes that concentricity requirements are noted on the motor datasheet, and should be considered when "sizing and selecting bearings for appropriate radial and preload forces to achieve desired motor running gap clearance and total runout". Bearings should also be chosen for the lowest possible friction and high quality lubricant, to reduce motor losses and heat generation. Similar Kollmorgen outrunner BLDC motors to the WK8115 can be observed to find appropriate concentricity requirements, and a factor of safety can be applied to account for the lower quality of the SteadyWin motor.

The Kollmorge

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- [2] Leslie Lamport (1994) *L^AT_EX: a document preparation system*, Addison Wesley, Massachusetts, 2nd ed.